

Fennel

Fennel (*Foeniculum vulgare*) is a flowering plant species in the carrot family.^[2] It is a hardy, perennial herb^[3] with yellow flowers and feathery leaves.^[4] It is indigenous to the shores of the Mediterranean but has become widely naturalized in many parts of the world, especially on dry soils near the sea-coast and on riverbanks.

It is a highly flavorful herb used in cooking and, along with the similar-tasting anise, is one of the primary ingredients of absinthe. **Florence fennel** or **finocchio** (UK: /fiˈnɒkioʊ/, US: /-ˈnoʊk-/, Italian: [fɪˈnɔkkjo]) is a selection with a swollen, bulb-like stem base that is used as a vegetable.

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Description

Foeniculum vulgare is a perennial herb. It is erect, glaucous green, and grows to heights of up to 2.5 metres (8 feet), with hollow stems. The leaves grow up to 40 centimetres (16 inches) long; they are finely dissected, with the ultimate segments filiform (threadlike), about 0.5 millimetres (1⁄64 in) wide. (Its leaves are similar to those of dill, but thinner.) The flowers are produced in terminal compound umbels 5–17.5 cm (2–7 in) wide,^[5] each umbel section having 20–50 tiny yellow flowers on short pedicels. The fruit is a dry schizocarp from 4–10 mm

Fennel

Fennel in flower
Scientific classification 
Kingdom: <u>Plantae</u>
Clade: <u>Tracheophytes</u>
Clade: <u>Angiosperms</u>
Clade: <u>Eudicots</u>
Clade: <u>Asterids</u>
Order: <u>Apiales</u>
Family: <u>Apiaceae</u>
Genus: <u>Foeniculum</u>
Species: <i>F. vulgare</i>
<u>Binomial name</u>
<i>Foeniculum vulgare</i> <u>Mill.</u>
<u>Synonyms</u> ^[1]
Synonymy ▪ <i>Anethum dulce</i> DC. ▪ <i>Anethum foeniculum</i> L. ▪ <i>Anethum minus</i> Gouan ▪ <i>Anethum panmori</i> Roxb.

($\frac{3}{16}$ – $\frac{3}{8}$ in) long, half as wide or less, and grooved.^[6] Since the seed in the fruit is attached to the pericarp, the whole fruit is often mistakenly called "seed".

- *Anethum panmorium* Roxb. ex Fleming
- *Anethum piperitum* Ucria
- *Anethum rupestre* Salisb.
- *Foeniculum azoricum* Mill.
- *Foeniculum capillaceum* Gilib.
- *Foeniculum divaricatum* Griseb.
- *Foeniculum dulce* Mill.
- *Foeniculum foeniculum* (L.) H.Karst.
- *Foeniculum giganteum* Lojac.
- *Foeniculum officinale* All.
- *Foeniculum panmorium* (Roxb.) DC.
- *Foeniculum piperitum* C.Presl
- *Foeniculum rigidum* Brot. ex Steud.
- *Ligusticum foeniculum* (L.) Roth
- *Ligusticum foeniculum* (L.) Crantz
- *Meum foeniculum* (L.) Spreng.
- *Meum piperitum* Schult.
- *Ozodia foeniculacea* Wight & Arn.
- *Selinum foeniculum* E.H.L.Krause
- *Seseli dulce* Koso-Pol.
- *Seseli foeniculum* Koso-Pol.
- *Seseli piperitum* Koso-Pol.
- *Tenoria romana* Schkuhr ex Spreng.



Florence fennel bulbs



Flower heads



Umbel



Fruits

Chemistry

The aromatic character of fennel fruits derives from volatile oils imparting mixed aromas, including trans-anethole and estragole (resembling liquorice), fenchone (mint and camphor), limonene,^[7] 1-octen-3-ol (mushroom).^[8] Other phytochemicals found in fennel fruits include polyphenols, such as rosmarinic acid and luteolin, among others in minor content.^[9]

F. vulgare essential oil also has non-food uses. Pavela et al 2016 find the EO to be insecticidal.^[10]

Similar species

Some plants in the Apiaceae family are poisonous and often difficult to identify.^[11]

Dill, coriander, ajwain, and caraway are similar-looking herbs, but shorter-growing than fennel, reaching only 40–60 cm (16–24 in). Dill has thread-like, feathery leaves and yellow flowers; coriander and caraway have white flowers and finely divided leaves (though not as fine as dill or fennel) and are also shorter-lived (being annual or biennial plants). The superficial similarity in appearance between these seeds may have led to a sharing of names and etymology, as in the case of meridian fennel, a term for caraway.

Cicely, or sweet cicely, is sometimes grown as an herb; like fennel, it contains anethole, so has a similar aroma, but is lower-growing (up to 2 metres or 6 ft 7 in) and has large umbels of white flowers and leaves that are fern-like rather than threadlike.

Giant fennel (*Ferula communis*) is a large, coarse plant, with a pungent aroma, which grows wild in the Mediterranean region and is only occasionally grown in gardens elsewhere. Other species of the genus *Ferula* are also commonly called giant fennel, but they are not culinary herbs.

In North America, fennel may be found growing in the same habitat and alongside natives osha (*Ligusticum porteri*) and *Lomatium* species, useful medicinal relatives in the parsley family.

Most *Lomatium* species have yellow flowers like fennel, but some are white flowered and resemble poison hemlock. *Lomatium* is an important historical food plant of Native Americans known as 'biscuit root'. Most *Lomatium* spp. have finely divided, hairlike leaves; their roots have a delicate rice-like odor, unlike the musty odor of hemlock. *Lomatium* species tend to prefer dry rocky soils devoid of organic material.

Etymology

Fennel came into Old English from Old French *fenoil* which in turn came from Latin *faeniculum*, a diminutive of *faenum*, meaning "hay".

Cultivation

Fennel is widely cultivated, both in its native range and elsewhere, for its edible, strongly flavored leaves and fruits. Its aniseed or liquorice flavor^[12] comes from anethole, an aromatic compound also found in anise and star anise, and its taste and aroma are similar to theirs, though usually not as strong.^[13]

Florence fennel (*Foeniculum vulgare* Azoricum Group; syn. *F. vulgare* var. *azoricum*) is a cultivar group with inflated leaf bases which form a bulb-like structure. It is of cultivated origin,^[14] and has a mild anise-like flavor, but is sweeter and more aromatic. Florence fennel plants are smaller than the wild type.^[15] Several cultivars of Florence fennel are also known by several other names, notably the Italian name *finocchio*. In North American supermarkets, it is often mislabeled as "anise".^{[16][17]}

Foeniculum vulgare 'Purpureum' or 'Nigra', "bronze-leaved" fennel, is widely available as a decorative garden plant.^[18]

Fennel has become naturalized along roadsides, in pastures, and in other open sites in many regions, including northern Europe, the United States, southern Canada, and much of Asia and Australia. It propagates well by both root crown and seed, and is considered an invasive species and a weed in Australia^[19] and the United States. It can drastically alter the composition and structure of many plant communities, including grasslands, coastal scrub, riparian, and wetland communities. It appears to do this by outcompeting native species for light, nutrients, and water and perhaps by exuding allelopathic substances that inhibit growth of other plants.^[20] In western North America, fennel can be found from the coastal and inland wildland-urban interface east into hill and mountain areas, excluding desert habitats.^{[21][22]} On Santa Cruz Island, California for example, fennel has achieved 50 to 90% absolute cover.^[20]

Production

As grouped by the United Nations Food and Agriculture Organization, production data for fennel are combined with similar spices – anise, star anise, and coriander.^[23] In 2014, India produced 60% of the world output of fennel, with China and Bulgaria as leading secondary producers (table).

Uses

Fennel was prized by the ancient Greeks and Romans who used it as medicine, food, and insect repellent. A fennel tea was believed to give courage to the warriors prior to battle. According to Greek mythology, Prometheus used a giant stalk of fennel to carry fire from Mount Olympus to Earth. Emperor Charlemagne required the cultivation of fennel on all imperial farms.^[24]

Florence fennel is one of the three main herbs used in the preparation of absinthe, an alcoholic mixture which originated as a medicinal elixir in Europe and became, by the late 19th century, a popular alcoholic drink in France and other countries.^[25] Fennel fruit is a common and traditional spice in flavored Scandinavian brännvin (a loosely defined group of distilled spirits, which include akvavit).^{[26][27]} Fennel is also featured in the Chinese Materia Medica for its medicinal functions.^[28]

Production of fennel – 2014

Country	Production (tonnes)
 <u>India</u>	584,000
 <u>China</u>	48,002
 <u>Bulgaria</u>	36,500
 <u>Iran</u>	32,771
 <u>Mexico</u>	29,251
 <u>Syria</u>	27,668
World	970,404

Data combined with related spices – anise, star anise & coriander. Source: FAOSTAT of the United Nations^[23]

Nutrition

A raw fennel bulb (235 g) consists of 212 g of water, 2.91 g of protein, 0.47 g of fat, and 17.2 g of carbohydrate (including 7.28 g of dietary fiber and 9.24 g of sugars), providing a total of 72.8 Calories (kcal) of energy. The 235g bulb provides 115 mg of calcium, 1.72 mg of iron, 40 mg of magnesium, 188 mg of phosphorus, 973 mg of potassium, 122 mg of sodium, trace amounts of zinc, copper, and selenium, 28.2 mg of vitamin C, as well as choline, several B vitamins, folate, beta-carotene, lutein, zeaxanthin, vitamin E, and vitamin K.^[29]



Sugar-coated and uncoated fennel fruits used as a breath freshener

Dried fennel fruits are typically used as a spice and normally eaten only in minute quantities. A 100-gram reference amount of fennel fruits provides 1,440 kilojoules (345 kilocalories) of food energy, and is a rich source (20% or more of the Daily Value, DV) of protein, dietary fiber, B vitamins and several dietary minerals, especially calcium, iron, magnesium and manganese, all of which exceed 100% DV (table). Fennel fruits are 52% carbohydrates (including 40% dietary fiber), 15% fat, 16% protein and 9% water (table).

Cuisine

The bulb, foliage, and fruits of the fennel plant are used in many of the culinary traditions of the world. The small flowers of wild fennel (known as fennel "pollen")^[30] are the most potent form of fennel, but also the most expensive.^[31] Dried fennel fruit is an aromatic, anise-flavored spice, brown or green in color when fresh, slowly turning a dull grey as the fruit ages. For cooking, green fruits are optimal.^[13] The leaves are delicately flavored and similar in shape to those of dill. The

Fennel seeds

Nutritional value per 100 g (3.5 oz)		
Energy	1,443 kJ (345 kcal)	
Carbohydrates	52 g	
<u>Dietary fiber</u>	40 g	
Fat	14.9 g	
<u>Saturated</u>	0.5 g	
<u>Monounsaturated</u>	9.9 g	
<u>Polyunsaturated</u>	1.7 g	
Protein	15.8 g	
Vitamins	Quantity	%DV[†]
<u>Thiamine (B₁)</u>	0.41 mg	36%
<u>Riboflavin (B₂)</u>	0.35 mg	29%
<u>Niacin (B₃)</u>	6.1 mg	41%
<u>Vitamin B₆</u>	0.47 mg	36%
<u>Vitamin C</u>	21 mg	25%
Minerals	Quantity	%DV[†]
<u>Calcium</u>	1196 mg	120%
<u>Iron</u>	18.5 mg	142%
<u>Magnesium</u>	385 mg	108%

bulb is a crisp vegetable that can be sautéed, stewed, braised, grilled, or eaten raw. Tender young leaves are used for garnishes, as a salad, to add flavor to salads, to flavor sauces to be served with puddings, and also in soups and fish sauce.^[32] Both the inflated leaf bases and the tender young shoots can be eaten like celery.^[12]

Fennel fruits are sometimes confused with those of anise, which are similar in taste and appearance, though smaller. Fennel is also used as a flavoring in some natural toothpastes. The fruits are used in cookery and sweet desserts.^[32]

Many cultures in India, Afghanistan, Iran, and the Middle East use fennel fruits in cooking. In Iraq, fennel seeds are used as an ingredient in nigella flavored breads.^[33] It is one of the most important spices in Kashmiri cuisine and Gujarati cooking.^[34] It is an essential ingredient of the Assamese/Bengali/Oriya spice mixture *panch phoron*^[35] and in Chinese five-spice powders. In many parts of India, roasted fennel fruits are consumed as *mukhwas*, an after-meal digestive and breath freshener (saunf), or candied as *comfit*. Fennel seeds are also often used as an ingredient in *paan*, a breath freshener most popularly consumed in India.^[33]

Fennel leaves are used in some parts of India as leafy green vegetables either by themselves or mixed with other vegetables, cooked to be served and consumed as part of a meal. In Syria and Lebanon, the young leaves are used to make a special kind of egg omelette (along with onions and flour) called *ijjeh*.

Many egg, fish, and other dishes employ fresh or dried fennel leaves. Florence fennel is a key ingredient in some Italian salads, or it can be braised and served as a warm side dish. It may be blanched or marinated, or cooked in risotto.

Fennel fruits are the primary flavor component in Italian sausage. In Spain, the stems of the fennel plant are used in the preparation of pickled eggplants, *berenjenas de Almagro*. An herbal tea or tisane can be made from fennel.

On account of its aromatic properties, fennel fruit forms one of the ingredients of the well-known compound liquorice powder. In the Indian subcontinent, fennel fruits are also eaten raw, sometimes with a sweetener.

In Israel, fennel salad is made of chopped fennel bulbs flavored with salt, black pepper, lemon juice, parsley, olive oil and sometimes sumac.

Culture

The Greek name for fennel is *marathon* (μάραθρον) or *marathos* (μάραθος),^[36] and the place of the famous battle of Marathon literally means a plain with fennel.^[37] The word is first attested in Mycenaean Linear B form as *ma-ra-tu-wo*.^[38] In Hesiod's *Theogony*, Prometheus steals the ember of fire from the gods in a hollow fennel stalk.^[39]

As Old English *finule*, fennel is one of the nine plants invoked in the pagan Anglo-Saxon *Nine Herbs Charm*, recorded in the 10th century.^[40]

<u>Manganese</u>	6.5 mg	310%
<u>Phosphorus</u>	487 mg	70%
<u>Potassium</u>	1694 mg	36%
<u>Sodium</u>	88 mg	6%
<u>Zinc</u>	4 mg	42%
Other constituents	Quantity	
Water	8.8 g	

[Link to Full USDA Database entry \(http://web.archive.org/web/20160312233538/https://ndb.nal.usda.gov/ndb/food/show/268?fg=&man=&facet=&count=&max=35&sort=&qlookup=Spices,+fennel+seed&offset=&format=Full&new=&measureby=\)](http://web.archive.org/web/20160312233538/https://ndb.nal.usda.gov/ndb/food/show/268?fg=&man=&facet=&count=&max=35&sort=&qlookup=Spices,+fennel+seed&offset=&format=Full&new=&measureby=)

Units

μg = micrograms • mg = milligrams

IU = International units

[†]Percentages are roughly approximated using US recommendations for adults.

In the 15th century, Portuguese settlers on Madeira noticed the abundance of wild fennel, and used the Portuguese word funcho (fennel) and the suffix -al to form the name of a new town, Funchal.^[41]

Henry Wadsworth Longfellow's 1842 poem "The Goblet of Life" (http://www.hwlongfellow.org/poems_poem.php?pid=75) repeatedly refers to the plant and mentions its purported ability to strengthen eyesight:

Above the lower plants it towers,
The Fennel with its yellow flowers;
And in an earlier age than ours
Was gifted with the wondrous powers
Lost vision to restore.



Fennel, from Köhler's Medicinal Plants (1887)

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External links

- Chisholm, Hugh, ed. (1911). "Fennel" (https://en.wikisource.org/wiki/1911_Encyclop%C3%A6dia_Britannica/Fennel). *Encyclopædia Britannica* (11th ed.). Cambridge University Press.
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